

“Li-Fi WORKS Despite Typical Jammers!”

Why is an Indian startup working on cutting-edge tech like Li-Fi? What advantages does Li-Fi hold over Wi-Fi? Who should invest in Li-Fi? EFY's Nitisha Dubey managed to speak to Deepak Solanki, Founder and CEO of Velmenni, to explore these questions and more...



DEEPAK SOLANKI
FOUNDER AND CEO, VELMENNI

How does your outdoor Li-Fi solution function?

Our LC Link consists of a transmitter and a receiver, with the transmitter being an LED. In front of the LED, we have a set of lenses that allow us to control the light's beam pattern. This setup enables us to place our system on a tower and input data through an Ethernet/fibre cable. A processor within the system modulates the Ethernet packet, which is then transmitted using the LED. The light passes through the lenses, focusing it on another tower, potentially up to one or two kilometres away. An identical system is placed on the second tower, focusing the transmitted light from the first to its receiver. The receiver then converts the light signal back into an electrical signal, decoding the data packets, including internet packets, and sending them back through an Ethernet/fibre cable. This process effectively connects the two mobile towers using our LC Link systems, with one transmitting the light and the other receiving and decoding it, thus enabling seamless data transfer between them.

What is the maximum effective distance of such a solution?

This long-range product can operate over distances of up to two kilometres. Within this two-kilometre radius, our LC Link can connect with multiple systems and mobile towers, allowing for seamless connectivity among several towers. The product is already available in the market, and we have successfully deployed it with multiple telecom operators in India and abroad. We have deployments in Sri Lanka, the US, and India, with ongoing trials and completed commercial implementations.

How does the Li-Fi system work, and what hardware is required to set it up in an office or home environment?

Our system is similar to Wi-Fi in terms of deployment. The Li-Fi access points are integrated with lighting fixtures and designed for office and home environments. We are now offering lights with built-in Li-Fi access points or routers. You can replace your existing light fixtures with our Li-Fi-enabled lights to use this technology. Similar to how you connect an Ethernet cable to a Wi-Fi

router, our Li-Fi lights require an Ethernet cable to carry data. Unlike Wi-Fi, Li-Fi is not built into phones or laptops by default. To access Li-Fi, we provide a small USB dongle that can be plugged into your computer. Once connected, when you position yourself under the Li-Fi light, you can access the internet through it. In summary, two pieces of hardware are required: a Li-Fi dongle, which plugs into your laptop, and a Li-Fi-enabled light source or Li-Fi router, which needs to be installed in your office or home. We handle the manufacturing of both these components.

How does Li-Fi compare to Wi-Fi in terms of power consumption and coverage?

Li-Fi is highly energy-efficient, using much less power than Wi-Fi for data transmission. By integrating an access point with existing office lighting, Li-Fi keeps power consumption below 10 watts while maintaining regular lighting functions. A single Li-Fi light, such as a 2x2 panel, can support up to 16 users, similar to how a Wi-Fi router connects multiple systems. This technology enables point-to-multiple communication,